

# Pedestrian Crossing Behavior in Interaction with Autonomous Vehicles : A VR Experimental Study

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**Abstract**—The integration of autonomous vehicles (AVs) into traffic road requires careful consideration of their ability to adapt to other road users. Among road users, pedestrians are particularly vulnerable and are likely to interact with AVs. While most studies have focused on pedestrian behavior in interactions with human-driven vehicles, the effects of AV presence and Vehicle-to-Pedestrian (V2P) communication on pedestrian behavior remain largely unexplored. This article explores the impact of these factors, as well as the influence of the learning process induced by the introduction of AVs into traffic, on pedestrian behavior. To this end, we analyze data from our experimental campaign involving 100 participants in a virtual reality (VR) environment designed to address this issue. This study provides a comprehensive analysis of pedestrian behavior, constituting a significant advancement in the understanding of pedestrian-autonomous vehicle interactions and contributing to the integration of AV.

**Index Terms**—Autonomous vehicle interaction, Pedestrian Behavior Analysis, V2P Communication

## I. INTRODUCTION

The latest innovations in the field of Autonomous Vehicles (AVs) have come about as a result of collecting and exploiting a large amount of driving data using AI technology. AVs are already being introduced in limited areas. Wider integration of AVs into road traffic is becoming increasingly realistic and imminent. The more AVs are used, the more their driving data can be leveraged to improve their control over the vehicle and their ability to adapt to their environment. This last point is essential, as the AV must adapt to a wide range of situations and environments, such as the type of interaction space and the type of road users (motorized vehicles, cyclists, pedestrians, etc.). Some agents, such as pedestrians, are particularly vulnerable to an AV, and therefore need to be the subject of in-depth studies.

In our previous articles [28], we focused on the use of Deep Reinforcement Learning (DRL) to design an AV model capable of interacting effectively with a pedestrian while ensuring their safety. With DRL, the AV control improves by experiencing interactions with the pedestrian. However, such a method requires a large number of scenarios, and it is hardly feasible to use real-world encounter scenarios without compromising pedestrian safety. Therefore, most learning solutions are based on the use of a simulated environment with a realistic pedestrian model, a model obtained through the use of data from studies on pedestrian behavior at crossings. The results

also indicate that better interaction efficiency can be achieved through vehicle control compared to classical human driving.

Classical studies of pedestrian behavior focus on the behavior of pedestrians based on data collected in uncontrolled interactions [14], [26]. Sensors are placed at frequent encounter points (crosswalks or intersections). The data is therefore representative of an interaction between a pedestrian and a vehicle controlled by a human. Such studies do not take into account the impact of the presence and behavior of an AV on the pedestrian. The AV can indeed be considered as a new type of road user with a specific behavior and communication.

The AV is a road user different from a traditional vehicle from the pedestrian's perspective for several reasons. First, the absence of a human driver, which is noticeable through the windshield, can alter the pedestrian's behavior. For example, they may take fewer risks due to concerns about the AV's control capabilities, or they may act more recklessly due to a lack of empathy toward the vehicle. Secondly, in order to facilitate the pedestrian's understanding of the vehicle's automation status [19], and to address the lack of direct communication between the driver and the pedestrian, studies have already proposed new communication solutions using external human-machine interfaces (eHMI), most of which are visual hardware-based supports.

Conducting experimental campaigns on pedestrian behavior in the presence of an AV is however particularly complex for two reasons. First, learning in a real-world environment is hardly feasible due to issues with replicability, complexity of implementation, and safety guarantees. Secondly, pedestrian behavior is also influenced by multiple factors: personal factors, those related to other agents in the environment, and those related to the interaction space. As a result, the generalization of pedestrian behavior model from behavioral studies is often quite limited, particularly due to the fixed encounter space in the experiments, but situational testing remains necessary (for the reasons mentioned above).

To address this issue, there are several experimental solutions, ranging from simple surveys and real-world tests to realistic virtual environments. We designed such an environment and conducted an experimental campaign with a hundred participants in collaboration with the CHU of Besançon. This study includes the use of a virtual pedestrian-AV encounter environment and carries out several encounter scenarios integrating V2P communication solutions. In this article, we will present a detailed analysis of pedestrian behavior in interaction with an AV.

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Our research focuses on the following aspects:

- Analyze the database: identify the characteristics of the participants and the specific features of the interaction.
- Explore the potential to classify pedestrian behavior in order to gain a deeper understanding of their actions.
- Study the key features of the pedestrian behavior: decision to cross, crossing speed and waiting time.
- Examine the impact of learning, implicit and explicit communication on pedestrian behavior and perception during the interaction.

## II. STATE OF THE ART

While many studies focus on the pedestrian behavior, a very few deal with interaction with an autonomous vehicle. Three types of pedestrian-related studies can be identified:

- Analytical studies of pedestrian behavior at a pedestrian crossing, typically based on recorded data from real-world crossings. Some articles also propose pedestrian behavior profiles or models.
- Studies on pedestrian behavior prediction, most often aimed at directly improving vehicle perception. They therefore adopt the vehicle's point of view and focus on the pedestrian's decision of crossing.
- Studies on V2P communication, propose various eHMI designs, evaluating and comparing their impact on pedestrian behavior. These studies aim to identify the relevance and the most effective communication solution, focusing on factors such as crossing time and overall pedestrian experience.

### A. Analytical studies

This type of study analyzes pedestrian behavior in relation to various environmental and personal influences.

1) *Risk perception*: Some studies evaluate pedestrians' perception and assessment of risk during a crossing. For example, article [8] analyzes the impact of pedestrians' perception of risk on their behavior, and the reasons for this perception. They discovered that this perception is not only linked to the behavior of the vehicle, but also to the encounter space in which the pedestrian evolves. Thus, the pedestrian's confidence is strongly impacted by the delimitation and visibility of this space. They also note a greater impact of distance from the vehicle compared to the vehicle speed. Moreover, this perception can have an impact on the pedestrian's behavior, with longer waiting times and higher speeds during the crossing.

2) *Behavioral factors analysis*: Other studies propose an analysis of the factors influencing pedestrian behavior. In this way, it is possible to detect the elements that impact his behavior, and in particular his decisions to cross or not. Important factors can be determined from a comparison of encounter scenarios, as in the article [20]. Personal factors appear, such as age [3], gender [14], gaze/body orientation [8] or experience (culture) [22], but also factors related to the environment, such as position and vehicle speed. In the

article [21], the authors look at the impact of time to collision (TTC) and vehicle speed/position on pedestrian choice, and also find that vehicle distance has a greater influence on pedestrian crossing choice than vehicle speed.

3) *Behavior profiles*: Some articles propose a classification of pedestrian behaviors or V2P interactions. For instance, article [16] evaluates the importance of different pedestrian characteristics, determining their impact on risk-taking, and also proposes a classification of pedestrians into 4 groups according to distinct behavioral characteristics. This distinction is based on personal characteristics such as age, gender, and experience. Another paper [15] classifies scenario types according to meta data such as collision risk, average agent speed, and intrusions (distance too close between two agents). In a different article, [26] separates pedestrians into two groups in the case of interaction at a crossing with a sign. One group includes the so-called opportune pedestrians (who seek to pass as quickly as possible by ensuring their safety), and the other includes pedestrians who respect the signalization communicated by the traffic lights. The categorization of pedestrians leads to a better understanding of pedestrian behavior (e.g. differences in trajectory). This last article then builds a specific pedestrian model by creating a state diagram.

### B. Predictive model studies

This type of study focuses on using pedestrian movement data to predict pedestrian behavior. Articles of this type usually propose the construction of a single pedestrian model by analyzing the characteristics of a pedestrian's movement in an encounter with a human-driven vehicle. Some offer image processing solutions to determine whether the pedestrian intends to cross [9], or exploit this data to directly develop decision-making or trajectory prediction models by integrating interaction-specific parameters.

1) *Pedestrian intention*: The first element to be analyzed with regard to pedestrian behavior is the detection of the pedestrian's intention to cross the road. In the article [10], the authors present a large number of articles on the subject. These mainly use databases of video clips (JAAD and PIE) of interaction between a vehicle and a pedestrian, and develop predictive tools to determine whether the pedestrian will enter the road. The aim is to predict the pedestrian's behavior from the vehicle's point of view, but these studies are limited to an encounter without an autonomous agent. This type of predictive solution is very complex, as it depends on multiple factors such as the pedestrian's personal characteristics, the environment, and the duration of the interaction. Another paper [21] shows that certain characteristics of the environment can compensate (if there is no information for one of the two) when predicting the pedestrian's intention (such as the study of the pedestrian's head and legs) but that analysis of the pedestrian's physical behavior is necessary to obtain an effective prediction.

2) *Critical Gap*: Another axis of research is concerned with the design of a model of pedestrian crossing behavior based

on the estimation of the pedestrian's critical gap (CG), i.e. a temporal space sufficient to ensure safe crossing from the pedestrian's point of view.

There are many articles on the subject. For instance, article [30] evaluates the pedestrian CG according to physical characteristics of the VP interaction (agents' trajectories). Another article [11] develops a CG estimation model based on personal (age, gender) and also environmental (relative position, speed) characteristics. Similarly, the article [24] evaluates CG according to the vehicle speed, the gap size, the pedestrian waiting time, and the frequency of attempts, then determines their respective impacts on CG prediction. [27] proposes an evaluation of the necessary parameters and a modeling of the CG based on data from a VR experiment.

These studies focus on providing a predictive model enabling the construction of an AV model capable of adapting to the pedestrian's behavior. However, this adaptation does not take into account the mutual impact of these two agents, nor the effect of the presence/communication of an AV in their evaluation. These elements can lead to possible changes in the pedestrian's behavior that are not properly identified in these models. In fact, the article [18] demonstrates that pedestrians can also communicate with other road users in a non-verbal manner. The autonomous vehicle can also be equipped with a communication tool. In this case, the AV must not only react to the pedestrian's actions but also be able to communicate and potentially cooperate with them. The autonomous vehicle can also have a communication tool.

### C. V2P communication studies

The last type of study addresses the issue of the mutual impact of the AV and the pedestrian. Several articles focus on designing an eHMI optimized to increase the pedestrian's comprehension, thereby increasing his confidence and safety during the interaction [7]. We identify two types of communication: implicit, which comes from the vehicle's behavior, and explicit, which manifests itself through the use of the eHMI. The article [23] highlights the greater importance of implicit communication compared with the explicit one, although its impact can vary across individuals. The authors also argue that explicit communication plays a crucial role in supporting and validating implicit communication.

Several types of eHMI are considered in the literature: either by the infrastructure (I2P), or directly by the vehicle (V2P). Several types of communication can be used, depending on the type of interaction. For example, [12] proposes a ground projection to warn of reversing vehicles in parking lots. V2P communication can be performed via a projection on the ground or via a strip light or 2D screen on the front of the vehicle. The article [7] also proposes a communication tool using cellphones, to overcome the problem of "smombies".

There are a large number of articles studying the impact of the presence of an eHMI and seeking to develop the most effective mode of communication:

- Impact of different types of communication on pedestrian behavior: The article [4] performs a study on the impact

of communication on pedestrian behavior using a virtual environment. Several parameters are taken into account: passive communication (type of vehicle), implicit communication (driving aggressiveness), and explicit communication (type and timing of eHMI with text or a simple strip light). A multiplicity of communication types is therefore studied and compared on crossing intention. In another article [25], the authors evaluate the impact of the eHMI and in particular its presence (without eHMI, with a static eHMI, with a mobile eHMI) according to three criteria of pedestrian behavior: crossing decisions, the pedestrian's gaze and the pedestrian experience. Some articles also study pedestrian behavior during a crossing according to their level of confidence [13], [4].

- Link between explicit and implicit communication: In the article [6], the authors compare the use of a strip light with several speed profiles. They note a cumulative impact of communication with the speed profile that is only present during cooperative vehicle behavior (slow deceleration), thus validating the article [23].
- Optimal eHMI: The articles [1] and [5] seek to determine the most effective type of eHMI, whether in terms of content (information on vehicle behavior or advice to pedestrians) or message form (symbol, text, colors).

### D. Pedestrian behavior study design

In the field of experimental pedestrian studies, it is very difficult to generalize results (i.e. separate pedestrian behaviors from their environmental context). This is due to the great diversity of the crossing spaces, such as the presence of signs, a variable number of agents (e.g. a group of pedestrians) or a lack of visibility. As highlighted by [10], this point requires the use of multiple databases and a large number of interactions.

The difficulty of designing experiments can also be a cause of error. For example, articles proposing studies on interaction with an autonomous vehicle can be faced with behavioral changes compared to a real crossing. According to [17], the construction of the study must be sufficiently intuitive, otherwise significant differences in behavior can be observed depending on the experimental procedure. If pedestrians are only asked to indicate the last moment when they feel confident to cross, without asking them to actually do it (stepping), we obtain a bias in the results with respect to a real crossing. The situation is therefore crucial for obtaining realistic results.

## III. PEDESTRIAN ANALYSIS

In our previous article [2], we defined a new experimental protocol inspired by VR studies of pedestrian behavior [17], [8], [29], [22] to build a realistic, replicable study that would yield valid results. To this end, we adapted the structure of the study, the nature of the data extracted, and the interaction scenarios. In particular, we have taken into account the limitations and areas for improvement proposed in the article [29]. Our study enables us to efficiently analyze the behavior of a pedestrian facing an autonomous vehicle, guaranteeing participant safety, replication of scenarios, and simplified

storage of behavioral data. Moreover, our database includes a range of variables, including personal information, responses to a post-experiment survey and behavioral measures under several test conditions.

We conduct the experiment in an industrial environment to simulate a realistic, enclosed space. This choice is based on several factors: such environments often restrict visibility, which helps direct attention to the interaction itself, and the deployment of autonomous vehicles is more feasible in these conditions, making such environments ideal for early-stage testing and observation.

In this experiment, participants were instructed to cross in four different types of scenarios to analyze V2P communication impact : Scenario *A* with adaptive speed, Scenario *S* with signaling, Scenario *N* without any communication, and Scenario *A&S* with both signaling and adaptive speed. To assess the impact of learning, each participant interacts with these types of AVs on the same scenario twice, once at the start of the experiment and once after experiencing random interaction scenarios.



Fig. 1: V2P communication in the VR environment

#### A. Database Analysis

In this first section, we analyze the characteristics of our participant group to identify potential biases and assess the generalizability of the study results. Although this study initially involved 100 participants, issues related to participant tracking in virtual space and data recording reduced the usable dataset to 81 participants minimum according to the data studied.

We identified several notable aspects concerning the sociodemographic characteristics of the participants. Information about the experience of the participants, defined as having lived or worked in different types of environment (urban, rural, and industrial), is also summarized in Table I. Data indicate that all participants are familiar with urban environments, a majority have experience in rural settings, and a significant proportion are familiar with industrial environments that involve forklift operations.

Regarding the participants' age, the distribution is not uniform, with an overrepresentation of young adults aged 24 to 26 years and a slight underrepresentation of individuals aged 33 to 39 years, as shown in Fig 2. On the contrary, the gender distribution is very balanced, with 43 women and 38 men.

TABLE I: Participant survey

| Category                              | 0     | 1     |
|---------------------------------------|-------|-------|
| Gender (0 men, 1 women)               | 46.9% | 53.1% |
| Urban (0 familiar, 1 unfamiliar)      | 100%  | 0%    |
| Rural (0 familiar, 1 unfamiliar)      | 81.5% | 18.5% |
| Industrial (0 familiar, 1 unfamiliar) | 39.5% | 60.5% |

With respect to the participants' experience with virtual reality, the majority (80%) reported having little to no prior exposure.

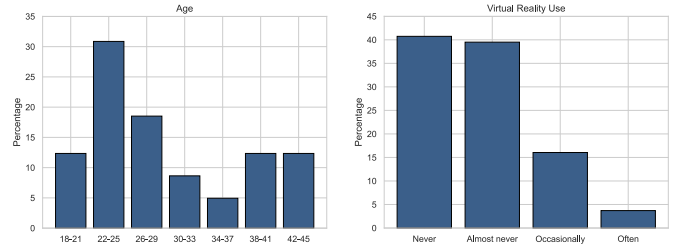


Fig. 2: Age and VR experience Distribution

The analysis of the correlation matrix of personal data reveals a slight relationship between experience in rural and industrial environments, another between industrial experience and the use of virtual reality, and a link between virtual reality use and gender.

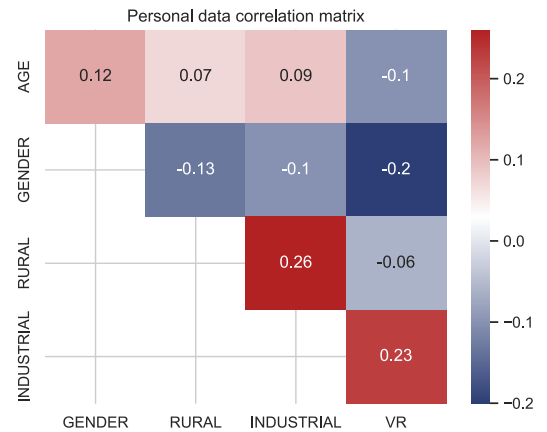


Fig. 3: Personal data correlation matrix

To summarize, our database is relatively well-balanced, the only bias found being the slight over-representation of young adults, and a cultural bias due to the fact that the experiment was conducted exclusively in Besançon (France).

#### B. Participant Perception

Following the encounter experiences with autonomous vehicles, each participant is asked to fill out a survey regarding their feelings about these different encounter tests. This allows us to gather each participant's perception of these encounters in terms of risk perception (**Q2**), decision to cross factor (**Q3**), and communication preferences/effectiveness (**Q1 / Q4**).

The post-experience survey included the questions :

- **Q1:** In which scenario did you feel most at ease?
- **Q2\_1:** How stressful did you find the A&S scenario?
- **Q2\_2:** How stressful did you find the A scenario?
- **Q2\_3:** How stressful did you find the S scenario?
- **Q2\_4:** How stressful did you find the N scenario?
- **Q3:** What did you base your crossing decision on?
- **Q4:** Did the signaling help you make the decision to cross?

TABLE II: Stress level survey

|                    | A & S | A    | S    | N    |
|--------------------|-------|------|------|------|
| Mean               | 0.52  | 1.30 | 1.58 | 2.38 |
| Standard Deviation | 0.94  | 1.18 | 1.20 | 1.37 |

Stress level with respect to the Likert scale (from 0 to 4)

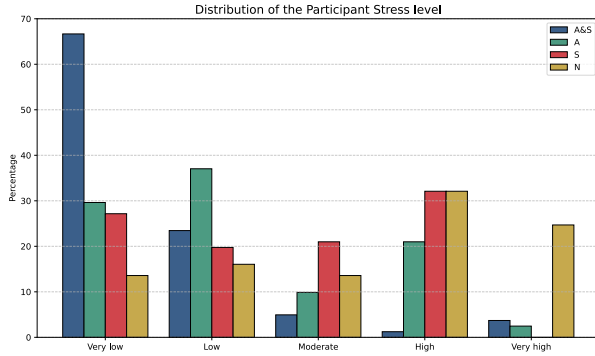


Fig. 4: Stress Level Distribution

Table II and Fig. 4 highlight a significant effect of communicative tools on participants' perceptions concerning stress level. Firstly, the introduction of communication tools generally leads to a reduction in stress, with adaptive speed (A) having a notably stronger effect. Additionally, these tools contribute to an improved consistency in stress levels, leading to a more uniform experience of stress among participants (preventing high-stress level experiences for specific participants). Secondly, the impact of communication tools is cumulative: the combination of speed adaptation and signaling yields more effective results, both in terms of the average and standard deviation, compared to using just one of these elements alone. In addition, the results of Q1 reveal that a large majority of participants preferred the A&S scenario, and around 65% consider that signaling has an impact on their decision making.

This cumulative impact can be explained by the behavior difference between participants according to the communicative tool used (one being more receptive than other on those specific communications). As illustrated in the Fig.5, the participants choose different factors to determine if they can cross or not, although the speed is the most selected feature. Participants who rely more on signaling are more confident in the S scenario than other participants (respectively the opposite with the A scenario) as we can see in Table III.

We also found no major correlation between the participant average stress level and personal data (Fig. 6), indicating that stress level is more influenced by the type of scenario rather than by personal information.

TABLE III: Stress level comparison with respect to Q3

| Group        | A & S | A    | S    | N    |
|--------------|-------|------|------|------|
| signaling    | 0.29  | 2.00 | 0.98 | 2.93 |
| no signaling | 0.57  | 1.15 | 1.72 | 2.27 |

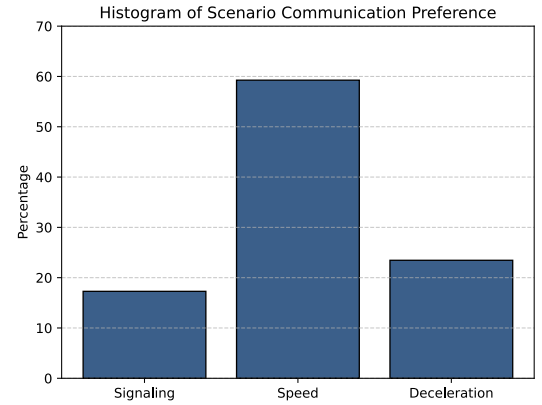


Fig. 5: Scenario Communication Preference

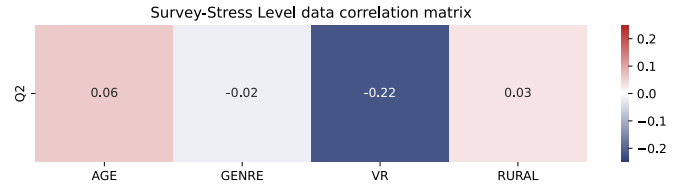


Fig. 6: Personal data - Stress Level Correlation

### C. Pedestrian crossing decision

Previous studies have evaluated pedestrians' critical gap (CG) based on socio-demographic characteristics such as age and gender. However, in our study, we did not find any strong correlation between these individual factors and CG. Additionally, we applied an XGBoost regression model to estimate the CG with those features, but it failed to produce a reliable prediction. These results suggest that the estimation of the critical gap is highly context-dependent and influenced by the specific circumstances of each interaction, making it difficult to predict based solely on participant scenario-invariant characteristics. The main reason for this failure is that the CG is strongly influenced by the pedestrian's decision to wait for the vehicle to stop before crossing.

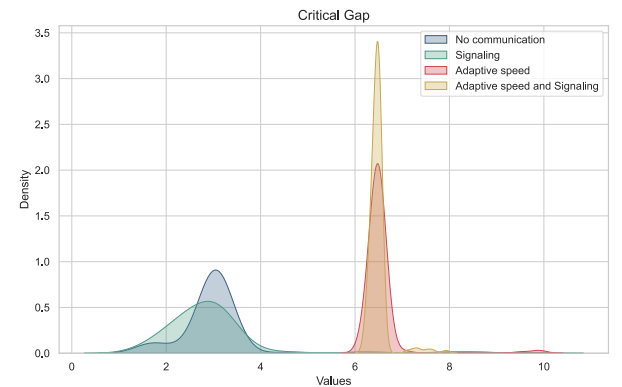


Fig. 7: Critical Gap Density by Scenario Type

In this study, we define the critical gap as the time obtained by dividing the relative distance between the pedestrian and the vehicle by the vehicle's speed at the start of the pedestrian crossing. If we consider only the CG when the pedestrian crosses before the vehicle stops, we observe a major difference linked to the integration of adaptive speed. As the vehicle slows down, the pedestrian crosses faster, with a higher CG as in Fig. 7. A total of 9 stops were observed in the scenario *S* compared to 15 stops in the scenario *S*. We can also note that the average crossing speed is 1.35m/s in our experiment, this speed is similar to that found in behavioral analyses in real environments. Initial analysis of the speed profiles during the scenarios and the average crossing speed indicates that the participants' behavior is very similar to real-world pedestrian actions, demonstrating a high degree of realism.

#### D. Learning process impact

The integration of new autonomous vehicles will have an impact on pedestrian behavior. He/She must learn how to interact with it. This learning process may modify the pedestrian's behavior towards the AV. To assess this impact, we will compare the results of each type of interaction scenario between discovery (first interaction) and post-experiment (after a learning phase). To do this, we extract features related to the interaction effectiveness: time lost by the vehicle and the pedestrian, critical gap, and pedestrian's waiting time. Table IV shows the variation in values across these variables.

TABLE IV: Differences in Mean and Standard Deviation of Efficiency Parameters (Post-experiment - Discovery)

|                | $\Delta\text{Mean}$ |           |           | $\Delta\text{Std}$ |           |           |
|----------------|---------------------|-----------|-----------|--------------------|-----------|-----------|
|                | $TA$                | $T_{p_p}$ | $T_{p_v}$ | $TA$               | $T_{p_p}$ | $T_{p_v}$ |
| <b>N</b>       | -19.47 %            | -33.21 %  | -32.36 %  | -11.03 %           | -10.85 %  | -10.47 %  |
| <b>S</b>       | -20.06 %            | -34.46 %  | -36.57 %  | -18.99 %           | -19.11 %  | -17.58 %  |
| <b>A</b>       | -22.05 %            | -41.81 %  | -31.53 %  | -49.92 %           | -48.39 %  | -51.07 %  |
| <b>A&amp;S</b> | -10.55 %            | -25.77 %  | -17.64 %  | -24.59 %           | -24.65 %  | -25.99 %  |

$TA$  : Pedestrian waiting time,  $Vm$  : Pedestrian speed,  
 $T_{p_v}$  et  $T_{p_p}$  : Vehicle and Pedestrian lost time,  
 Vehicle behavior: Nominal N, Adaptive speed A,  
 Signalisation S, Adaptive speed and signalisation A&S.

A general improvement in encounter effectiveness was observed, highlighting the importance of pedestrian learning in shaping their behavior when facing an autonomous vehicle. The ranking of scenario types in terms of performance remained unchanged; however, the improvement was more pronounced in the non-A&S scenarios. This may be due to the potential for participants to adjust their behavior in these cases (less instinctive scenarios). The variability of efficiency-related variables was also significantly reduced, particularly for the A scenario. This element shows the participants' ability to adapt and learn how to interact effectively with the vehicle depending on its communication.

#### E. Pedestrian Behavior Clustering

As presented in the related work section, some studies focus on clustering pedestrians according to their crossing behavior. This allows us to identify and get a better understanding of their behavior. After applying a PCA model to the vehicle

and pedestrian motion data, the distribution of pedestrians in the 2D space is represented in Fig. 8. The optimal number

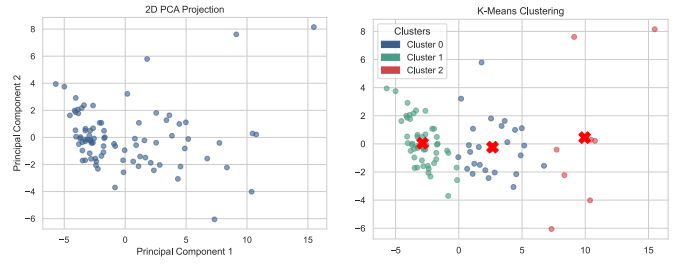


Fig. 8: PCA and Kmeans

of classes is difficult to determine based on the BIC analysis, the silhouette score, or the elbow method. Indeed, we observe no significant distinction in pedestrian behaviour, which limits any clear classification of pedestrians. However, by fixing the number of clusters to three and applying k-means clustering, we find that pedestrians are mainly clustered based on their behavior in the scenarios *S* and *N*, due to a higher variance.

The analysis of the mean factors for each cluster provides a better understanding of this clustering. The variables with the greatest impact on the clustering of pedestrians (from the *S* and *N* scenarios) are the time lost and the vehicle information when the pedestrian crosses, while personal information such as age and gender is not significant.

Three types of behavior can be identified, corresponding to the three resulting classes. In cluster 2, participants tend to be cautious pedestrians who wait for the vehicle to come to a complete stop (the vehicle's speed is close to zero before crossing, and their critical gap is very high). This cluster, however, displays high variance and a low number of participants, suggesting that it includes extreme cases where interactions have been particularly ineffective. In cluster 1, participants quickly cross before the vehicle arrives, assessing their ability to cross based on the vehicle's speed (vehicle speed remains near the limit prior to the crossing, and waiting time is very short). Lastly, in cluster 0, participants wait only for the vehicle to communicate (the vehicle's speed decreases before crossing, and they report a much higher average stress level than participants in other clusters in the *N* scenario). This cluster is not clearly defined by a specific crossing decision strategy; participants wait only briefly before crossing and appear to be influenced by vehicle communication such as in *S* scenario, and even more in A scenario.

We note that adding the survey data does not impact the clustering. We also try to classify participants based on their responses to Question 3, which reflects their perception of the reasons behind their decision to cross. In this clustering, the variance within the groups is too high to clearly distinguish them. While a direct link between the previous and current classifications is not possible, some concordances are observed in Fig. 9, for instance most participants in cluster 2 belong to the *signaling* group and most participants in cluster 1 belong to the *'vehicle speed'* group. Our findings also show that



participants who relied on the vehicle's speed performed better than those who relied on signaling or deceleration.

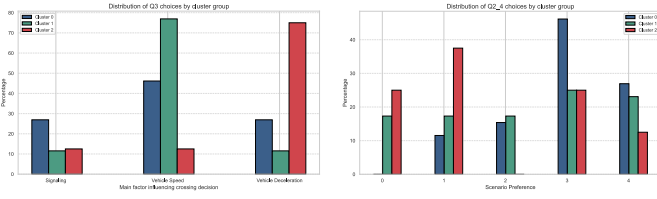


Fig. 9: Survey choices per cluster group

#### F. V2P Communication Impact

We aim to examine the impact of vehicle communication on participant behavior. To analyze these effects, we compare key variables in interaction efficiency (such as waiting time, lost time for pedestrians and vehicles) for each type of scenario. This analysis method is similar to the approach used in section III-D. The Table V shows a global improvement of the

TABLE V: Scenario key-features

| Mean | Discovery |        |        | Post-Learning |        |        |
|------|-----------|--------|--------|---------------|--------|--------|
|      | $TA$      | $Tl_p$ | $Tl_v$ | $TA$          | $Tl_p$ | $Tl_v$ |
| N    | 6.06      | 3.89   | 4.36   | 4.78          | 2.49   | 2.84   |
| S    | 5.57      | 3.39   | 3.73   | 4.47          | 2.23   | 2.41   |
| A    | 4.87      | 2.72   | 3.42   | 3.83          | 1.59   | 2.36   |
| A&S  | 4.08      | 1.97   | 2.64   | 3.60          | 1.41   | 2.13   |

$TA$  : Pedestrian waiting time (s),  
 $Tl_v$  /  $Tl_p$  : Vehicle and Pedestrian lost time (s)  
 Vehicle behavior: Nominal N, Adaptive speed A,  
 Signalisation S, Adaptive speed and signalisation A&S.

interaction efficiency related to the addition of communication tools. However, indirect (adaptive speed) and direct (signaling) communications have distinct effects on the interaction. First, adaptive speed communication leads to better average interaction efficiency, as indicated by improved values for both  $TA$  and  $Tl_p$  (scenario A). Second, the reduction in uncertainty of pedestrian behavior, reflected by a decrease in the standard deviation, is even more pronounced with the introduction of signaling communication (scenario S) for the pedestrian features ( $TA$  and  $Tl_p$ ). It is also noted that the time lost by the vehicle is largely influenced by the indirect communication. Anticipated deceleration, due to the adaptive speed communication, helps reduce the time lost by enabling the vehicle to achieve a higher speed at the end of the pedestrian crossing and to reach its cruising speed more quickly after the interaction.

We also observe the same cumulative effect as in the section III-B, whether considering the mean value or the standard deviation (Table VI). Indeed, participants are not equally affected by acceleration and signaling (Fig. 9). This effect can be explained by variations in pedestrian behavior according to the categories of individuals. Lastly, we note that enhancing pedestrians' understanding and confidence level leads to a slight decrease in their average speed during crossing.

TABLE VI: Percentage Differences Relative to Scenario N

| Mean | $TA$     | $v$      | $Tp_v$   | $Tp_p$   |
|------|----------|----------|----------|----------|
| S    | -6.03 %  | -1.77 %  | -14.43 % | -10.06 % |
| A    | -19.35 % | -2.34 %  | -15.51 % | -35.64 % |
| A&S  | -23.98 % | -4.51 %  | -23.57 % | -42.73 % |
| Std  | $TA$     | $v$      | $Tp_v$   | $Tp_p$   |
| S    | -15.85 % | -1.24 %  | -14.14 % | -14.79 % |
| A    | -3.90 %  | 1.23 %   | -23.19 % | -5.61 %  |
| A&S  | -32.39 % | -16.60 % | -47.32 % | -33.60 % |

We aim to evaluate the stability of pedestrian classification across different scenarios. As identified in Section III-E, the global classification highlighted that the most influential variables were associated with the scenario N. This scenario features the most contrasted pedestrian behaviors, making it particularly suitable for classification. After splitting the scenarios, we apply clustering to the dataset for scenario N.

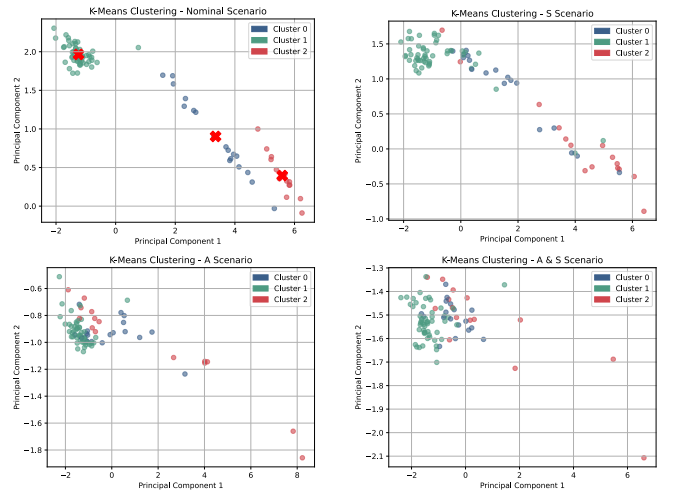


Fig. 10: Clustering based on N Scenario

TABLE VII: Silhouette score

|       | N    | S    | A    | A & S |
|-------|------|------|------|-------|
| Score | 0.58 | 0.18 | 0.13 | 0.11  |

Based on Fig. 10, we analyze their behavior in other scenarios. The objective was to verify whether certain behavioral profiles identified initially would remain consistent across different contexts (i.e., whether behavioral profiles could be preserved independently of environmental changes). We observe a decrease in classification relevance in the scenarios S and A. This effect becomes even more pronounced in scenario A&S. To quantify this loss of classification discriminability, the table VII presents the silhouette score for each type of scenario, illustrating the reduction in cluster separation as pedestrian behaviors become more consistent. This uniformity of pedestrian behavior is a very important point, as it enables better prediction on the part of the vehicle, thus ensuring greater safety and time efficiency.

#### IV. CONCLUSION

In this article, we study pedestrian behavior during interaction with an autonomous vehicle. To this end, we analyze the results of an experimental study, involving around 100 participants evolving in a VR environment as pedestrians facing different AV behaviors. This study focuses on a specific interaction, limited to a particular industrial environment and involving a single interaction between an AV and a pedestrian. Based on a wide-ranging database in terms of personal information as well as crossing behaviors, our study examines how V2P communication strategies and the learning process influence pedestrian behavior.

Results reveal that pedestrian behavior is strongly influenced by the vehicle's communication. In particular, variations in the critical gap across different scenarios demonstrate the limitations in existing critical gap modeling approaches. The results indicate a global improvement in the quality of interactions, both in terms of efficiency and pedestrian comfort, following the introduction of V2P communication strategies. Moreover, a cumulative effect of these communication tools was identified. A significant impact of the learning process is also highlighted, leading to enhanced scenario efficiency and faster adaptation to a new agent, particularly in communicative scenarios.

After the pedestrian behavior clustering (primarily based on CG), specific patterns in their decision-making were identified. Three distinct groups emerged, relying respectively on the vehicle's deceleration, the vehicle's speed, or the complete stop of the vehicle to initiate crossing.

These findings provide a better understanding of pedestrian behavior when interacting with an AV. This research leads the way for future developments in the simulation of realistic pedestrian agents, incorporating various behavioral profiles (particularly with regard to decision-making) to improve the relevance of simulations involving AV control models.

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